

CS5707 Artificial Intelligence
Probabilistic Inference

Yongmin Li

Overview

- **Uncertainty**
- **Basic probability theory**
- **Bayesian reasoning – Naïve Bayes**
- **Bayesian networks**

Reading

[1] M. Negnevitsky (2011). §3 Uncertainty management in rule-based systems.

[2] Russell, S. and Norvig, P. (2021). §12, Quantifying uncertainty.

[3] Russell, S. and Norvig, P. (2021). §13, Probabilistic reasoning.

Uncertainty

Uncertainty

- Information can be incomplete, inconsistent, uncertain → often unsuitable for solving a problem.
- **Uncertainty:** the lack of the exact knowledge that would enable us to reach a perfectly reliable conclusion.
- Classical logic: ***law of the excluded middle***

IF ***A*** **is true**
THEN ***A*** **is not false**

IF ***A*** **is false**
THEN ***A*** **is not true**

Sources of Uncertain Knowledge

- **Weak implications**
- **Imprecise languages**
- **Unknown data**
- **Combining views of different experts**

Weak Implications

- Domain experts and knowledge engineers have the painful task of establishing **concrete** correlations between IF (condition) and THEN (action) parts of the rules.
- Intelligent systems need to have the ability to handle vague associations, for example by accepting the degree of correlations as **numerical certainty factors**.

Imprecise Language

- Our natural language is ambiguous and imprecise.
- Examples: words as
  - *often*
  - *sometimes*
  - *frequently*
  - *hardly ever*.
- It is difficult to express knowledge in the precise form
- ==> Quantify the meaning of the facts

Quantification of ambiguous and imprecise terms on a time-frequency scale

Ray Simpson (1944)		*Milton Hakel* (1968)	
Term	*Mean value*	*Term*	*Mean value*
Always	99	Always	100
Very often	88	Very often	87
Usually	85	Usually	79
Often	78	Often	74
Generally	78	Rather often	74
Frequently	73	Frequently	72
Rather often	65	Generally	72
About as often as not	50	About as often as not	50
Now and then	20	Now and then	34
Sometimes	20	Sometimes	29
Occasionally	20	Occasionally	28
Once in a while	15	Once in a while	22
Not often	13	Not often	16
Usually not	10	Usually not	16
Seldom	10	Seldom	9
Hardly ever	7	Hardly ever	8
Very seldom	6	Very seldom	7
Rarely	5	Rarely	5
Almost never	3	Almost never	2
Never	0	Never	0

Unknown Data

Incomplete or missing data

- to accept the value “unknown” &
- to proceed with approximate reasoning

Combining Views of Different Experts

- Large expert systems usually combine the knowledge and expertise of a number of experts.
- Unfortunately, experts often have contradictory opinions and produce conflicting rules.

How to resolve the conflict?

- Assign a weight to each expert and then calculate the composite conclusion.
- No systematic method exists to obtain these weights.

Basic Probability Theory

- Theory of probability: 17th century
- Concept: long history

  "probably", "likely", "maybe", "perhaps" and "possibly" in our spoken languages
- **Probability**
  - the proportion of cases in which the event occurs
  - a scientific measure of chance.

Probability Theory

Probability (1)

- Probability: 0 <= P <= 1
  0: an absolute impossibility
  1: an absolute certainty
- An example
  Probability of success/failure

$$P(success) = \frac{the\ number\ of\ \ successes}{the\ number\ of\ \ possible\ outcomes}$$

$$P(\ failure) = \frac{the\ number\ of\ \ failures}{the\ number\ of\ \ possible\ outcomes}$$

Probability (2)

- If s is the number of times success can occur, and f is the number of times failure can occur, then

$$P(success) = p = \frac{s}{s+f}$$

$$P(failure) = q = \frac{f}{s+f}$$

and $$p + q = 1$$

Conditional Probability (1)

- Two events: *A* and *B*
- Occur conditionally on the occurrence of the other

Conditional probability:

- the probability that event *B* will occur if event *A* occurs.
- *p*(*B*|*A*): *Conditional probability of event A occurring given that event B has occurred*

$$p(B|A) = \frac{\text{the number of times both } A \text{ and } B \text{ can occur}}{\text{the number of times } A \text{ can occur}} = \frac{p(B \cap A)}{p(A)}$$

Conditional Probability (2)

Joint probability: the probability that **both** A and B will occur.
Represented as $p(A \cap B)$ or $p(AB)$, $p(A,B)$.

$$p(A|B) = \frac{p(A \cap B)}{p(B)}$$

Similarly, the conditional probability of event B occurring given that event A has occurred equals

$$p(B|A) = \frac{p(B \cap A)}{p(A)}$$

Hence,

$$p(A \cap B) = p(A|B) \times p(B)$$
$$p(B \cap A) = p(B|A) \times p(A)$$

This leads to

$$p(A|B) \times p(B) = p(B|A) \times p(A)$$

and yields the **Bayesian rule**:

$$p(B|A) = \frac{p(A|B) \times p(B)}{p(A)}$$

Bayesian Rule

$$p(B|A) = \frac{p(A|B) \times p(B)}{p(A)}$$

where:

$p(A|B)$ is the conditional probability that event A occurs given that event B has occurred;

$p(B|A)$ is the conditional probability of event B occurring given that event A has occurred;

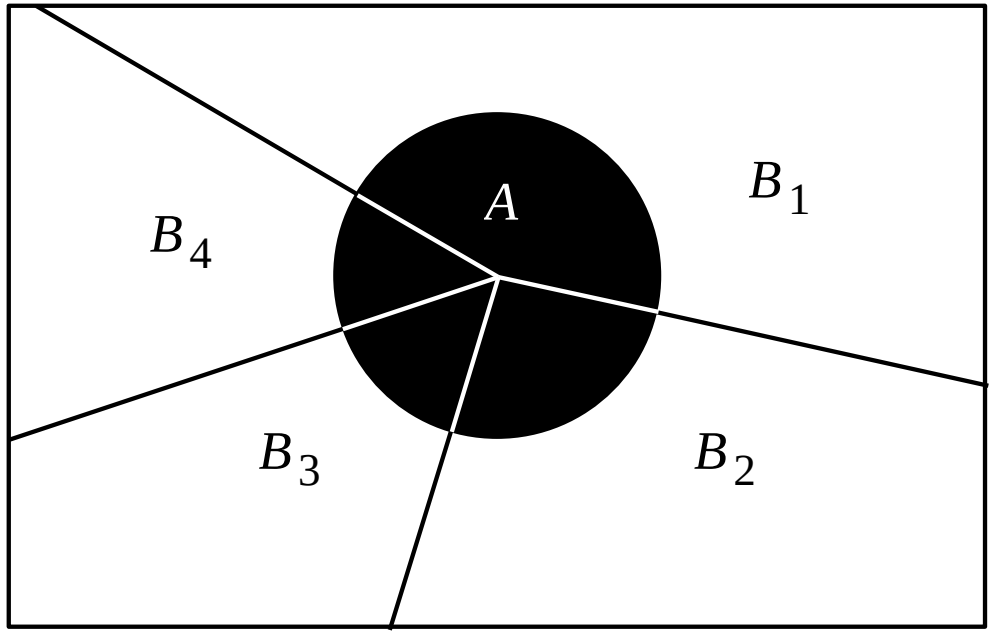
$p(A)$ is the probability of event A occurring;

$p(B)$ is the probability of event B occurring.

Marginalisation from Joint Probabilities

- Often *p(A)* is not given directly
- But we can sum it out from $p(A \cap B_i)$ when
- B_i are (1) mutually exclusive; and (2) exhaustive

$$p(A) = \sum_{i=1}^{n} p(A \cap B_i) = \sum_{i=1}^{n} p(A|B_i) \times p(B_i)$$



Bayesian Reasoning

Bayesian Reasoning (1)

A Simple Case:

If the occurrence of event E depends on only two mutually exclusive events, H and NOT H, we obtain:

$$p(H|E) = \frac{p(E|H) \times p(H)}{p(E|H) \times p(H) + p(E|\neg H) \times p(\neg H)}$$

Bayesian Reasoning (2)

Suppose all rules in the knowledge base are represented in the following form:

IF *H* is true
THEN *E* is true {with probability *p*}

This rule implies that if hypothesis *H* occurs, then the probability that event *E* will occur is *p*.

In expert systems, *H* usually represents a hypothesis and *E* denotes evidence to support this hypothesis.

Bayesian Reasoning (3)

$$p(H|E) = \frac{p(E|H) \times p(H)}{p(E|H) \times p(H) + p(E|\neg H) \times p(\neg H)}$$

where:

$p(H)$ is the prior probability of hypothesis H being true;

$p(E|H)$ is the probability that hypothesis H being true will result in evidence E;

$p(\neg H)$ is the prior probability of hypothesis H being false;

$p(E|\neg H)$ is the probability of finding evidence E even when hypothesis H is false.

Bayesian Reasoning (4)

- An expert determines the **prior probabilities** $p(H)$ and $p(\neg H)$ **conditional probabilities** $p(E|H)$ and $p(E|\neg H)$.
- Users provide information about the evidence observed E
- The expert system computes $p(H|E)$ for hypothesis H in light of the user-supplied evidence E.
- Probability $p(H|E)$ is called the **posterior probability** of hypothesis H upon observing evidence E.

Bayesian Reasoning (5)

- **The General Case**:
  multiple hypotheses $H_1, H_2, \ldots, H_m$ and
  multiple evidences $E_1, E_2, \ldots, E_n$.
- Single evidence E and multiple hypotheses follow:

$$p(H_i|E) = \frac{p(E|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E|H_k) \times p(H_k)}$$

- Multiple evidences and multiple hypotheses follow:

$$p(H_i|E_1\, E_2 \ldots E_n) = \frac{p(E_1\, E_2 \ldots E_n|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E_1\, E_2 \ldots E_n|H_k) \times p(H_k)}$$

Bayesian Reasoning (6)

- Problems in computing
  $p(E_1, E_2, \ldots, E_n | H_i)$
- Conditional independence among different evidences assumed.
  $$p(\mathrm{E}_1, \mathrm{E}_2, \ldots, \mathrm{E}_n | H_i) = p(\mathrm{E}_1 | H_i) \times p(\mathrm{E}_2 | H_i) \times \ldots \times p(\mathrm{E}_n | H_i) \times p(H_i)$$
- Therefore
  $$p(H_i | \mathrm{E}_1\ \mathrm{E}_2 \ldots \mathrm{E}_n)$$
  $$= \frac{p(\mathrm{E}_1 | H_i) \times p(\mathrm{E}_2 | H_i) \times \ldots \times p(\mathrm{E}_n | H_i) \times p(H_i)}{\sum_{k=1}^{m} p(\mathrm{E}_1 | \mathrm{H}_k) \times p(\mathrm{E}_2 | \mathrm{H}_k) \times \ldots \times p(\mathrm{E}_n | H_k) \times p(H_k)}$$

Bayesian Reasoning (7)

- This reasoning model is called **Naïve Bayes** for the assumption of Conditional independence of evidences given the hypothesis
- The above assumption is not always true!
- But Naïve Bayes works surprisingly well in practice!

An Example

Suppose an expert, given three conditionally independent evidences E_1, E_2 and E_3, creates three mutually exclusive and exhaustive hypotheses H_1, H_2 and H_3, and provides prior probabilities for these hypotheses – $p(H_1)$, $p(H_2)$ and $p(H_3)$, respectively. The expert also determines the conditional probabilities of observing each evidence for all possible hypotheses.

The Prior and Conditional Probabilities

Probability	*Hypothesis*		
	$i = 1$	$i = 2$	$i = 3$
$p(H_i)$	0.40	0.35	0.25
$p(E_1 \mid H_i)$	0.3	0.8	0.5
$p(E_2 \mid H_i)$	0.9	0.0	0.7
$p(E_3 \mid H_i)$	0.6	0.7	0.9

- **Without any evidence, belief in H_1 is the highest.**

After Evidence E3 Observed

$$p(H_i|E_3) = \frac{p(E_3|H_i) \times p(H_i)}{\sum_{k=1}^{3} p(E_3|H_k) \times p(H_k)}, \qquad i = 1, 2, 3$$

Probability	Hypothesis		
	$i = 1$	$i = 2$	$i = 3$
$p(H_i)$	0.40	0.35	0.25
$p(E_1\|H_i)$	0.3	0.8	0.5
$p(E_2\|H_i)$	0.9	0.0	0.7
$p(E_3\|H_i)$	0.6	0.7	0.9

Thus,

$$p(H_1|E_3) = \frac{0.6 \cdot 0.40}{0.6 \cdot 0.40 + 0.7 \cdot 0.35 + 0.9 \cdot 0.25} = 0.34$$

$$p(H_2|E_3) = \frac{0.7 \cdot 0.35}{0.6 \cdot 0.40 + 0.7 \cdot 0.35 + 0.9 \cdot 0.25} = 0.34$$

$$p(H_3|E_3) = \frac{0.9 \cdot 0.25}{0.6 \cdot 0.40 + 0.7 \cdot 0.35 + 0.9 \cdot 0.25} = 0.32$$

- **Belief in H_2 increases and becomes equal to belief in H_1.**
- **Belief in H_3 also increases and nearly reaches beliefs in H_1 and H_2.**

Suppose We Observe Evidence E1 Now

$$p(H_i|E_1E_3) = \frac{p(E_1|H_i) \times p(E_3|H_i) \times p(H_i)}{\sum_{k=1}^{3} p(E_1|H_k) \times p(E_3|H_k) \times p(H_k)}, \qquad i = 1, 2, 3$$

Probability	Hypothesis		
	$i = 1$	$i = 2$	$i = 3$
$p(H_i)$	0.40	0.35	0.25
$p(E_1\|H_i)$	0.3	0.8	0.5
$p(E_2\|H_i)$	0.9	0.0	0.7
$p(E_3\|H_i)$	0.6	0.7	0.9

Hence,

$$p(H_1|E_1E_3) = \frac{0.3 \cdot 0.6 \cdot 0.40}{0.3 \cdot 0.6 \cdot 0.40 + 0.8 \cdot 0.7 \cdot 0.35 + 0.5 \cdot 0.9 \cdot 0.25} = 0.19$$

$$p(H_2|E_1E_3) = \frac{0.8 \cdot 0.7 \cdot 0.35}{0.3 \cdot 0.6 \cdot 0.40 + 0.8 \cdot 0.7 \cdot 0.35 + 0.5 \cdot 0.9 \cdot 0.25} = 0.52$$

$$p(H_3|E_1E_3) = \frac{0.5 \cdot 0.9 \cdot 0.25}{0.3 \cdot 0.6 \cdot 0.40 + 0.8 \cdot 0.7 \cdot 0.35 + 0.5 \cdot 0.9 \cdot 0.25} = 0.29$$

Hypothesis H_2 has now become the most likely one.

Finally We Observe Evidence E2

$$p(H_i|E_1E_2E_3) = \frac{p(E_1|H_i) \times p(E_2|H_i) \times p(E_3|H_i) \times p(H_i)}{\sum_{k=1}^{3} p(E_1|H_k) \times p(E_2|H_k) \times p(E_3|H_k) \times p(H_k)}, \qquad i = 1, 2, 3$$

Probability	Hypothesis		
	$i = 1$	$i = 2$	$i = 3$
$p(H_i)$	0.40	0.35	0.25
$p(E_1\|H_i)$	0.3	0.8	0.5
$p(E_2\|H_i)$	0.9	0.0	0.7
$p(E_3\|H_i)$	0.6	0.7	0.9

$$p(H_1|E_1E_2E_3) = \frac{0.3 \cdot 0.9 \cdot 0.6 \cdot 0.40}{0.3 \cdot 0.9 \cdot 0.6 \cdot 0.40 + 0.8 \cdot 0.0 \cdot 0.7 \cdot 0.35 + 0.5 \cdot 0.7 \cdot 0.9 \cdot 0.25} = 0.45$$

$$p(H_2|E_1E_2E_3) = \frac{0.8 \cdot 0.0 \cdot 0.7 \cdot 0.35}{0.3 \cdot 0.9 \cdot 0.6 \cdot 0.40 + 0.8 \cdot 0.0 \cdot 0.7 \cdot 0.35 + 0.5 \cdot 0.7 \cdot 0.9 \cdot 0.25} = 0$$

$$p(H_3|E_1E_2E_3) = \frac{0.5 \cdot 0.7 \cdot 0.9 \cdot 0.25}{0.3 \cdot 0.9 \cdot 0.6 \cdot 0.40 + 0.8 \cdot 0.0 \cdot 0.7 \cdot 0.35 + 0.5 \cdot 0.7 \cdot 0.9 \cdot 0.25} = 0.55$$

Although the initial ranking was H_1, H_2 and H_3, only hypotheses H_1 and H_3 remain under consideration after all evidences (E_1, E_2 and E_3) were observed.

Bias of the Bayesian Method

- Humans cannot elicit probability values consistently with the Bayesian rules or do it badly.
- This suggests that the conditional probabilities may be inconsistent with the prior probabilities given by the expert.

An Example: Consistence of an Expert's Judgement (1)

Consider a car that does not start and makes odd noises when you press the starter.

OddNoises: the symptom is "odd noises"
StarterFaulty: the starter if faulty.

An expert's judgement:

$p(\text{StarterFaulty}|\text{OddsNoises}) = 0.70$
$p(\text{StarterFaulty}) = 0.05$

Empirical studies:

$p(\text{OddNoises}|\text{StarterFaulty}) = 0.85$

$p(\text{OddNoises}|\neg\ \text{StarterFaulty}) = 0.20$

Given the empirical data, is the expert's judgement consistent?

An Example (2)

- We can apply the Bayesian rule to obtain:

$$p(StarterFaulty \mid OddNoises)$$

$$= \frac{p(OddNoises \mid StarterFaulty) \bullet p(StarterFaulty)}{p(OddNoises \mid StarterFaulty) \bullet p(StarterFaulty) + p(OddNoises \mid \neg StarterFaulty) \bullet p(\neg StarterFaulty)}$$

$$= \frac{0.85 \cdot 0.05}{0.85 \cdot 0.05 + 0.20 \cdot 0.95} = 0.18$$

- **The number obtained is significantly lower than the expert's estimate of 0.7!**

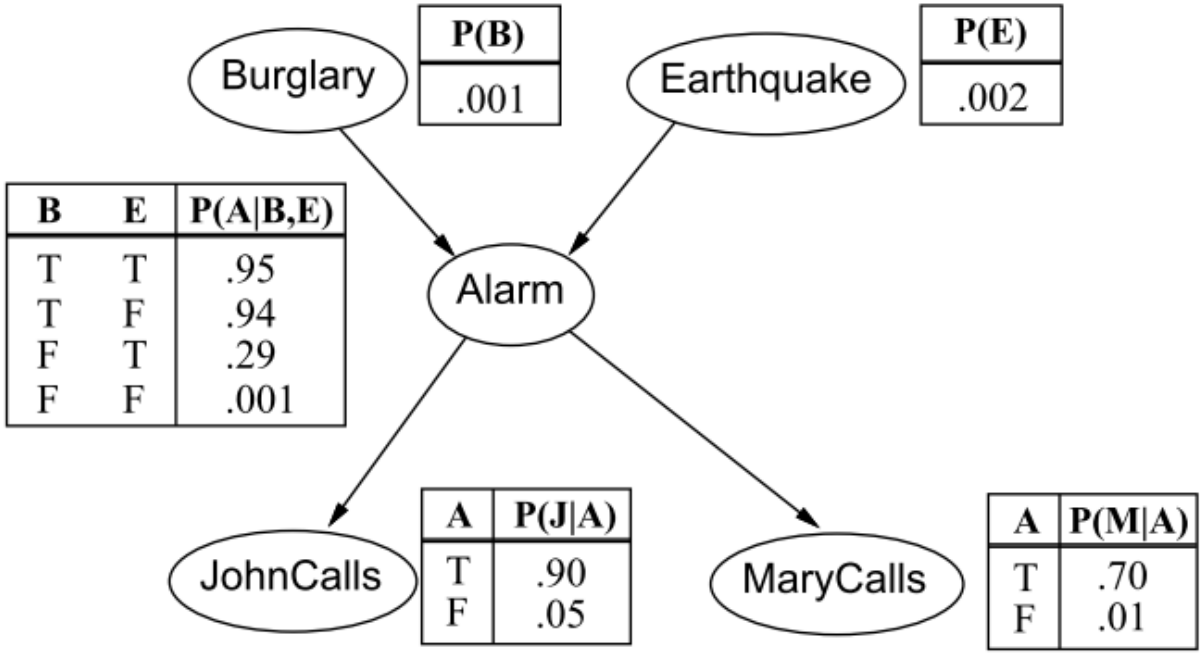
Bayesian Networks

Bayesian Networks

A simple graphical notational independence assertions and hence for compact specification of full joint distributions.

- A set of nodes: variables
- A directed, acyclic graph
- A conditional distribution for each node given its parents – the simplest case:
  conditional probability table (CPT)

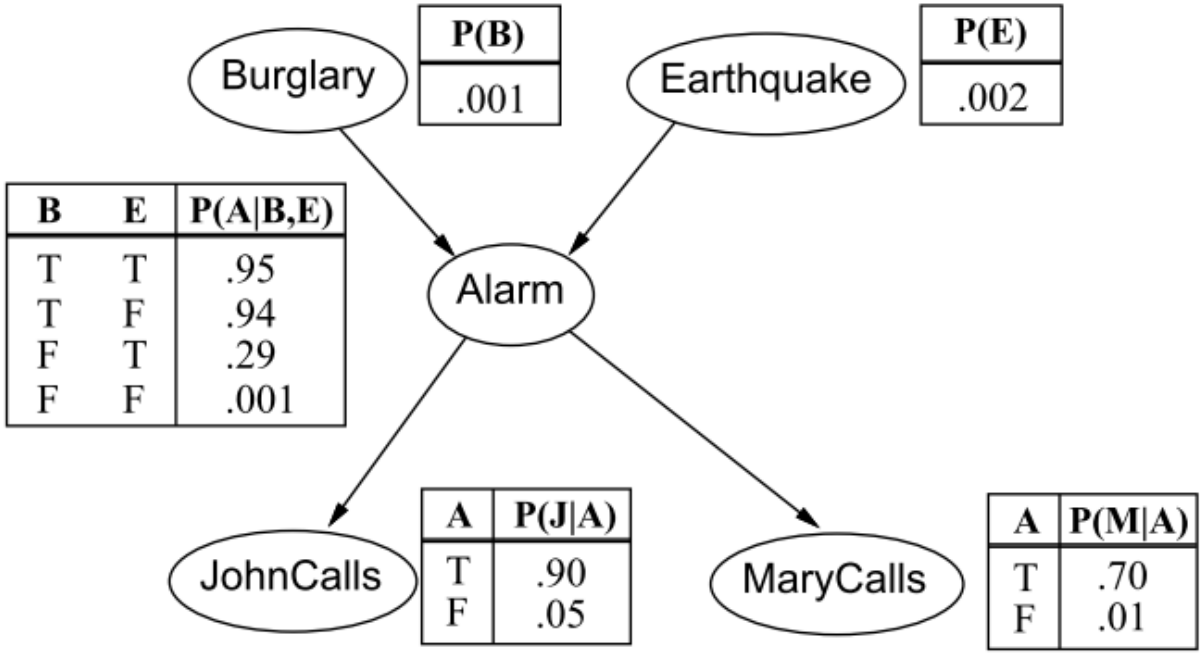
Example of Bayesian Networks



The network reflects causal knowledge:

- ✓ A burglar can set the alarm off
- ✓ An earthquake can set the alarm off
- ✓ The alarm can cause John to call
- ✓ The alarm can cause Mary to call

Full Joint Distribution



Full Joint Distribution = product of local conditional distributions

Example:

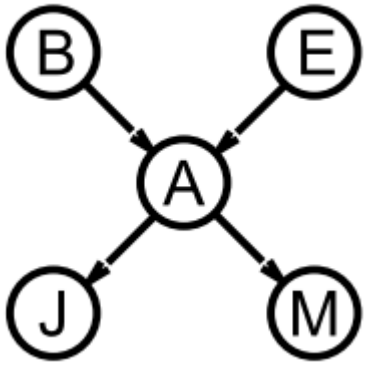
$$P(j, m, a, \neg b, \neg e) = P(j \mid a)P(m \mid a)P(a \mid \neg b \wedge \neg e)P(\neg b)P(\neg e)$$
$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 = 0.000628 .$$

Inference in Bayesian Networks

- Exact inference by enumeration
- Exact inference by variable elimination
- Approximate inference by stochastic simulation
- Approximate inference by Markov Chain Monte Carlo (MCMC)

Inference by Enumeration

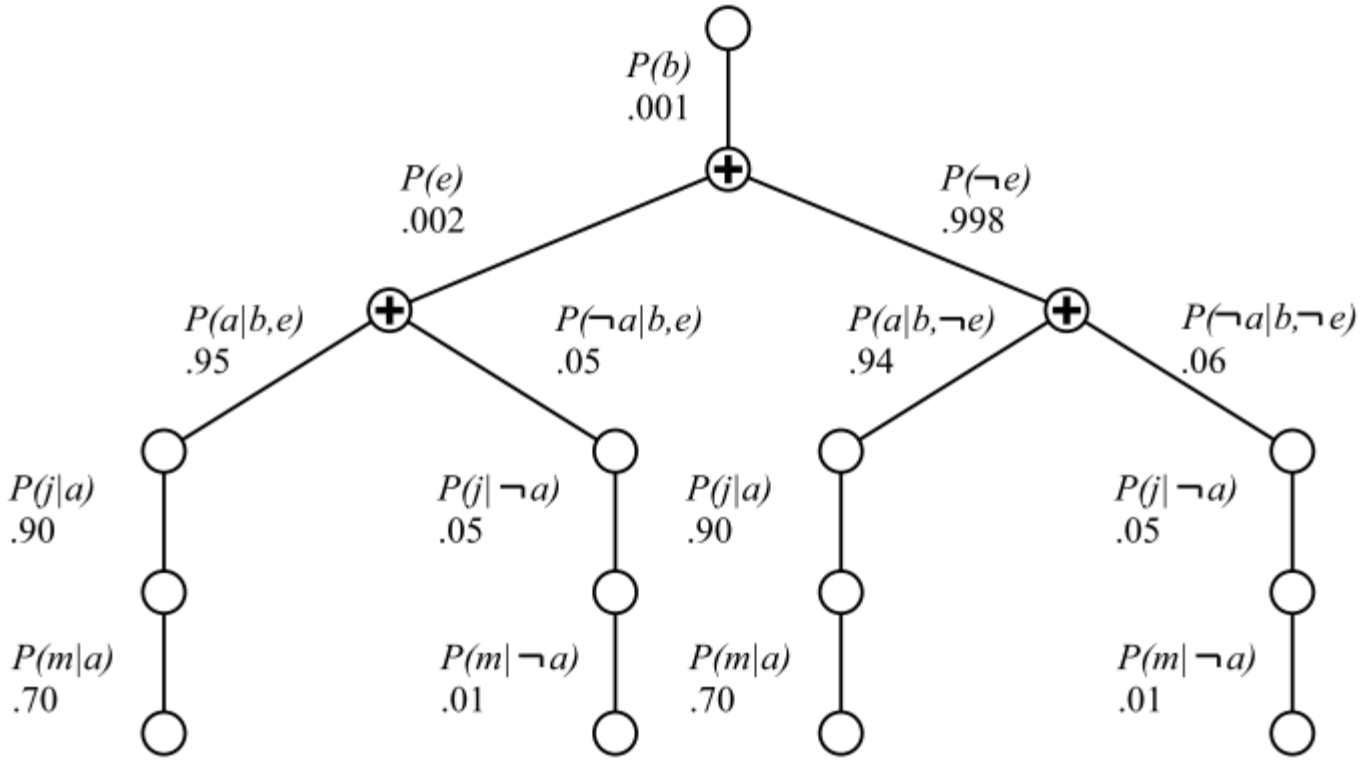
$$\begin{aligned}
&\mathbf{P}(B|j,m) \\
&= \mathbf{P}(B,j,m)/P(j,m) \\
&= \alpha\mathbf{P}(B,j,m) \\
&= \alpha \sum_e \sum_a \mathbf{P}(B,e,a,j,m)
\end{aligned}$$



Evaluate the full joint entries using CPTs:

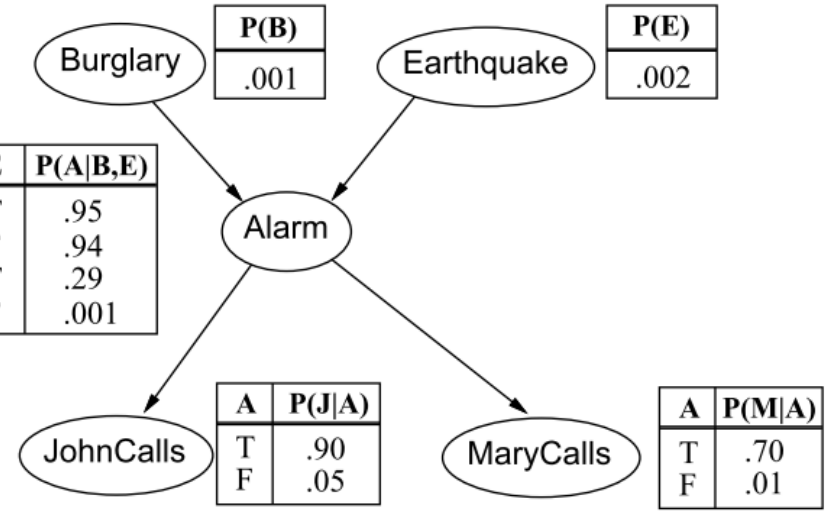
$$\begin{aligned}
&\mathbf{P}(B|j,m) \\
&= \alpha \sum_e \sum_a \mathbf{P}(B)P(e)\mathbf{P}(a|B,e)P(j|a)P(m|a) \\
&= \alpha\mathbf{P}(B) \sum_e P(e) \sum_a \mathbf{P}(a|B,e)P(j|a)P(m|a)
\end{aligned}$$

Inference by Enumeration

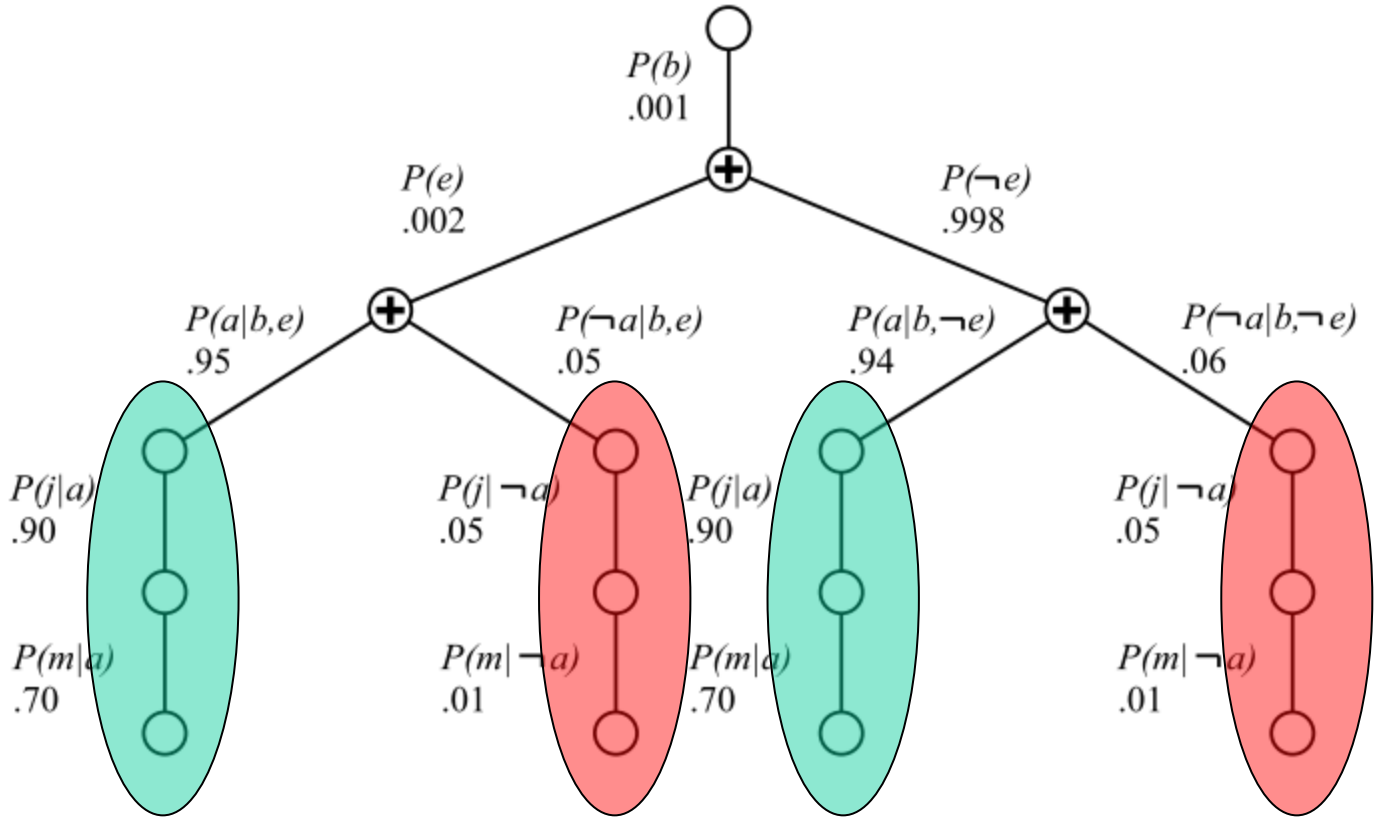


$$\mathbf{P}(B|j,m) = \alpha \langle 0.00059224, 0.0014919 \rangle \approx \langle 0.284, 0.716 \rangle .$$

$$\begin{aligned}
&\mathbf{P}(B|j,m) \\
&= \alpha \textstyle\sum_e \sum_a \mathbf{P}(B)P(e)\mathbf{P}(a|B,e)P(j|a)P(m|a) \\
&= \alpha \mathbf{P}(B) \textstyle\sum_e P(e) \sum_a \mathbf{P}(a|B,e)P(j|a)P(m|a)
\end{aligned}$$



Inference by Enumeration

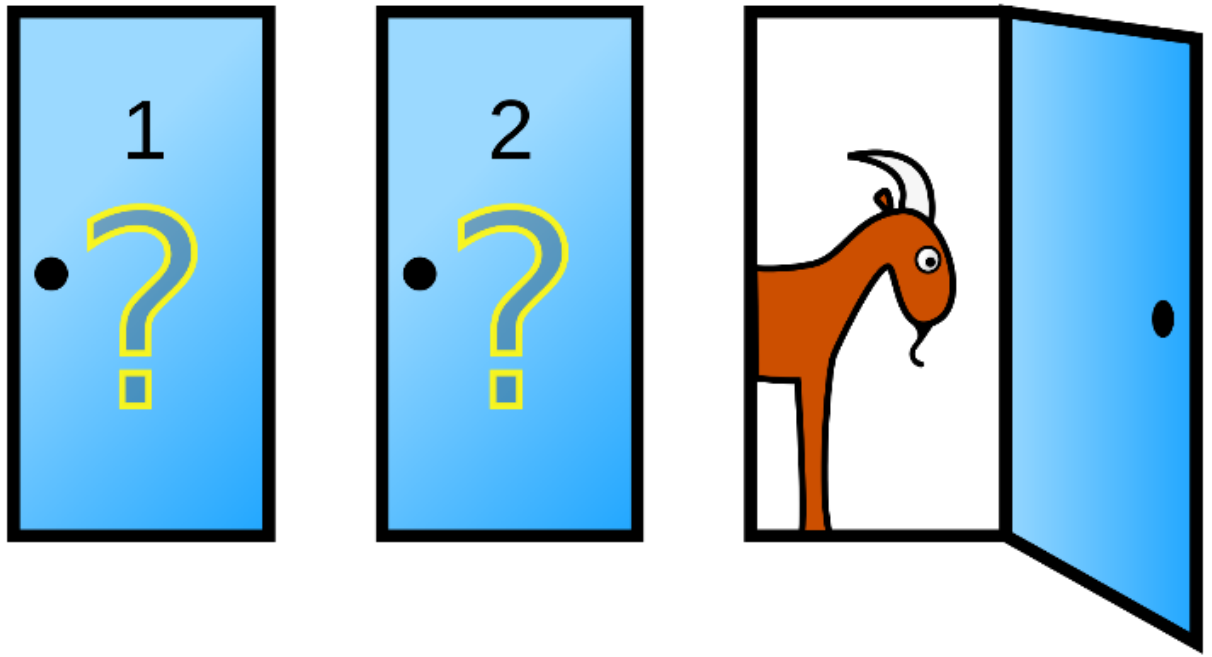


Enumeration is inefficient: repeated computation!

Inference by Variable Elimination

- The same principle as in enumeration
- Tricks to reduce computation:
  - sum out variables from a product of factors
  - move constants outside
  - add up sub-matrices in pointwise product of remaining factors

Monty Hall Problem



Behind three closed doors, there is a car (the prize) and two goats (worthless). You pick a door, say No. 1. Then the host (Monty Hall in the show), who knows what are behind the doors, opens another door that has a goat, say No. 3.

Now you have the chance to decide to stick to your original choice of No. 1, or to switch to No. 2.

Summary

- **Uncertainty**
- **Basic probability theory**
- **Bayesian reasoning – Naïve Bayes**
- **Bayesian networks**